

BDIC – BE IoT Project No. 1

Staff Member:	Anshu Mukherjee
Programme:	BE IoT
No. Students:	3
Project Title:	Dual-Mode Reconfigurable Intelligent Surface for Simultaneous Wireless Information and Power Transfer

Project Description:

Reconfigurable intelligent surfaces (RIS) have emerged as a key enabling technology for sixth generation (6G) wireless networks, offering low-cost and energy-efficient control over wireless propagation environments. By deploying arrays of programmable reflecting elements, RIS can intelligently manipulate electromagnetic waves to enhance signal coverage, improve spectral efficiency, and reduce interference. However, conventional RIS operating purely in reflecting mode suffers from the "multiplicative fading" effect, where the cascaded transmitter-RIS-receiver channel experiences significant path loss that limits achievable performance gains, particularly when the RIS is deployed far from both the base station and users. RIS operating entirely in amplifying mode addresses this limitation by integrating power amplifiers that boost reflected signals, effectively compensating for path loss. However, this comes at the cost of increased power consumption, hardware complexity, thermal noise, and higher implementation costs. A dual-mode architecture that strategically combines reflecting and amplifying elements offers a compelling middle ground, balancing the coverage extension benefits of amplification against energy efficiency considerations.

Simultaneously, simultaneous wireless information and power transfer (SWIPT) has attracted considerable research interest as a paradigm for enabling self-sustainable Internet of Things devices and sensor networks through radio frequency energy harvesting. Integrating dual-mode RIS with SWIPT systems presents an opportunity to jointly optimize information throughput and harvested energy while maintaining practical power budgets. This project investigates such integration in a downlink multi-user system, where the RIS operates in a dual-mode configuration: a subset of elements functions in reflecting mode providing phase-controlled reflection, while the remaining elements operate in amplifying mode capable of simultaneous phase control and signal amplification. The student will develop a mathematical system model incorporating amplifier noise, power consumption constraints, and nonlinear energy harvesting characteristics. The core optimization problem involves jointly designing the base station beamformer and RIS coefficients to maximize spectral efficiency subject to energy harvesting requirements, or to characterize the rate-energy trade-off region. An alternating optimization framework using successive convex approximation will be employed, with the mode ratio studied parametrically through simulations.

Suits students who like:

- Mathematical modelling and algorithm development for wireless communication systems
- Working with linear algebra, matrix operations, and optimization theory
- Convex optimization techniques and iterative algorithm design
- Programming in MATLAB or Python
- Exploring emerging 6G technologies, green communications, and sustainable wireless networks
- Signal processing concepts including beamforming and MIMO systems

Project Overview:

The student will begin by conducting a comprehensive literature review on RIS-aided communications, dual-mode architectures, and SWIPT system design. Following this, the student will develop a mathematical system model for a dual-mode RIS-aided SWIPT downlink scenario, deriving expressions for signal-to-interference-plus-noise ratio at information receivers and harvested power at energy harvesting receivers. Practical considerations including amplifier noise figures, maximum amplification constraints, and nonlinear energy harvesting circuit characteristics will be incorporated.

The core technical work involves formulating a joint optimization problem over the base station transmit beamformer and RIS coefficients for both reflecting-mode and amplifying-mode elements. Given the non-convex nature of this problem, the student will design an alternating optimization algorithm, iteratively solving beamforming and RIS coefficient subproblems using successive convex approximation or semidefinite relaxation techniques. Convergence behaviour and computational complexity will be analysed.

Performance evaluation will be conducted through extensive MATLAB or Python simulations, comparing the proposed dual-mode scheme against purely reflecting RIS, purely amplifying RIS, and baseline systems without RIS. Key performance metrics include achievable sum-rate, total harvested energy, rate-energy trade-off regions, and overall energy efficiency. The impact of system parameters such as the number of RIS elements, mode ratio between reflecting and amplifying elements, amplification power budget, and user locations will be systematically investigated. Deliverables include a complete simulation framework, technical documentation, and a final report with potential for conference publication if results demonstrate sufficient novelty and performance gains.

BDIC – BE IoT Project No. 2

Staff Member:	Anshu Mukherjee
Programme:	BE IoT
No. Students:	3
Project Title:	Optimization of Secret Key Rate in UAV-HAP Communications using Rician Channel Model for Physical Layer Security

Project Description:

The increasing demand for reliable and secure wireless communications has led to the exploration of High-Altitude Platforms (HAPs) as a promising solution for future generation communication networks. This research project focuses on optimizing the secret key rate between Unmanned Aerial Vehicles (UAVs) and HAPs, addressing the critical challenge of secure communication in aerial networks. Students will develop simulation models using MATLAB or Python to investigate the physical layer security aspects of UAV-HAP communications. The project will analyze various factors affecting the secret key generation process, including atmospheric conditions, mobility patterns, channel characteristics, and potential eavesdropping scenarios. By implementing different optimization algorithms, students will work to maximize the achievable secrecy rate while maintaining communication reliability and minimizing the impact of potential security threats.

Suits students who like:

- Communications and Signal Processing
- Programming and Simulation
- Mathematical Modelling
- Security and Cryptography
- Emerging Technologies

Project Overview:

The project implementation will progress through several key phases. Initially, students will develop a Rician channel model for UAV-HAP communications, incorporating both line-of-sight (LOS) and scattered components. The Rician K-factor analysis will consider the dynamic nature of aerial communications and environmental impacts. Students will then derive achievable secrecy rate expressions considering the Rician fading channels between the UAV-HAP link and potential eavesdropper links. The optimization phase will focus on maximizing the secrecy rate through various algorithms, considering practical constraints such as UAV energy limitations and HAP coverage requirements.

The simulation phase will involve implementing the theoretical models and optimization algorithms in MATLAB or Python to evaluate system performance under different scenarios. Students will analyze the impact of various Rician K-factors, eavesdropper locations, and UAV mobility patterns on the achievable secrecy rate. The project will conclude with a comprehensive analysis of results,

including secrecy rate performance metrics, outage probability analysis, and comparative studies with existing security schemes.

Related work:

[1] Y. Zhang, Z. Na, W. Lu and C. Ji, "One Friendly UAV Against Multiple Mobile Eves in HAP Networks: A Novel Physical Layer Secure Communication Scheme," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 1, pp. 1352-1364, Jan. 2025, doi: 10.1109/TVT.2024.3451508.

[2] Azizi, Arman, Mustafa Kishk, and Arman Farhang. "Exploring the Impact of HAPS-RIS on UAV-based Networks: A Novel Architectural Approach." *arXiv preprint arXiv:2409.17817* (2024).

BDIC – BE IoT Project No. 3

Staff Member:	Anshu Mukherjee
Programme:	BE IoT
No. Students:	3
Project Title:	Deep Reinforcement Learning for Secure STAR-RIS Networks with Cross-Side Eavesdropping

Project Description:

Simultaneous transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) has emerged as a promising technology for sixth generation (6G) wireless networks, enabling full-space coverage by allowing each surface element to simultaneously transmit and reflect incident signals. However, this architecture introduces a unique security challenge: users on one side of the STAR-RIS inherently receive signal components intended for users on the opposite side. This creates a cross-side eavesdropping scenario where transmission-side users become potential eavesdroppers for reflection-side communications and vice versa. Unlike external eavesdroppers who receive weak leakage signals, these internal users receive intentionally strong signals, fundamentally complicating physical layer security design.

This project investigates secure beamforming for STAR-RIS assisted multi-user systems under cross-side eavesdropping using deep reinforcement learning (DRL). The system comprises a multi-antenna base station serving users on both sides of a STAR-RIS, where each user group is treated as potential eavesdroppers for the opposite group. The objective is to maximise sum secrecy rate while satisfying quality-of-service requirements and STAR-RIS energy conservation constraints. This presents unique challenges: improving signal quality for one group may enhance their eavesdropping capability against the other, creating complex security-fairness-rate trade-offs that traditional optimisation methods struggle to handle.

The student will develop a DRL framework using Actor-Critic algorithms such as Deep Deterministic Policy Gradient to learn adaptive beamforming policies mapping channel states to joint base station precoding and STAR-RIS coefficient configurations. Performance evaluation will compare against conventional STAR-RIS without security considerations, reflection-only RIS baselines, and classical optimisation methods.

Suits students who like:

- Machine learning and reinforcement learning implementation
- Physical layer security and information-theoretic concepts
- Wireless communication system simulation
- Programming in Python (PyTorch, Stable-Baselines3) or MATLAB
- Exploring emerging 6G technologies and intelligent surfaces

Project Overview:

The student will begin with a literature review on STAR-RIS fundamentals, physical layer security, and DRL techniques. The implementation phase involves developing a multi-user STAR-RIS simulator with realistic channel models, formulating cross-side eavesdropping scenarios, and deriving secrecy rate expressions for each user. The core work involves formulating secrecy rate maximisation as a Markov Decision Process, where the state includes channel information, actions comprise beamforming vectors and STAR-RIS coefficients, and rewards balance secrecy rates against cross-side leakage. The student will train DRL agents and evaluate convergence, secrecy performance, and fairness between user groups.

Comparative analysis will benchmark against non-secure STAR-RIS, reflection-only RIS, and classical methods. Deliverables include a simulation framework, trained models, and a final report with potential for conference publication.

References:

- [1] Y. Liu et al., "STAR-RIS: Simultaneous Transmitting and Reflecting Reconfigurable Intelligent Surface," *IEEE Commun. Lett.*, vol. 25, no. 9, pp. 3134–3138, Sep. 2021.
- [2] X. Li et al., "Enhancing Secrecy Performance for STAR-RIS NOMA Networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 2684–2688, Feb. 2023.
- [3] C. Huang, R. Mo, and C. Yuen, "Reconfigurable Intelligent Surface Assisted Multi-User MISO Systems Exploiting Deep Reinforcement Learning," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1839–1850, Aug. 2020.

BDIC – BE IoT Project No. 4

Staff Member:	Dr. Alzubair Hassan
Programme:	BE IoT
No. Students:	4
Project Title:	Design and Implementation of a Privacy-Preserving Digital Forensic System for Encrypted Data

Project Description:

Modern software systems increasingly rely on encryption to protect sensitive data and ensure user privacy. While encryption is essential for security, it presents significant challenges for digital forensics, where investigators require access to evidence for incident response, auditing, and compliance. Rather than attempting to bypass or weaken encryption, this project requires the student to design and implement a secure software system that enables forensic analysis while preserving data confidentiality.

In this project, the student will design a digital forensic system architecture that supports controlled forensic analysis over encrypted data sources such as system logs, application data, network traces, or IoT data streams. The system must be engineered using secure software engineering principles, including security-by-design, least privilege, separation of duties, and auditability.

The proposed system should incorporate components such as encrypted data storage, secure key management, forensic access control, and privacy-aware analysis modules. Instead of full decryption, the system will support techniques such as metadata analysis, cryptographic hashing, integrity verification, selective decryption, or secure logging to extract useful forensic evidence without exposing sensitive content. All forensic actions must be logged and auditable to prevent misuse or evidence tampering.

Students are expected to perform threat modelling to identify risks (e.g., key leakage, insider abuse, evidence manipulation) and to design appropriate mitigation mechanisms. The system will be evaluated in terms of security, privacy preservation, forensic usefulness, and performance overhead, demonstrating how forensic readiness can be engineered into modern encrypted software systems.

Related Work:

Malik, A. and Sharma, A.K., 2023. Blockchain-based digital chain of custody multimedia evidence preservation framework for internet-of-things. *Journal of Information Security and Applications*, 77, p.103579.

Loffi, L., Camillo, G.L., De Souza, C.A., Westphall, C.M. and Westphall, C.B., 2025. Management of the Chain of Custody of Digital Evidence Using Blockchain and Self-Sovereign Identities: A Systematic Literature Review. *IEEE Access*.

BDIC – BE IoT Project No. 5

Staff Member:	Dr. Alzubair Hassan
Programme:	BE IoT
No. Students:	4
Project Title:	A Federated Intrusion Detection Mechanism with Adaptive Privacy for UAV Swarms

Project Description:

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are increasingly deployed across application domains such as urban surveillance, emergency response, environmental monitoring, and precision agriculture. Their ability to operate autonomously, collect real-time data, and collaborate as swarms makes them highly effective in complex and distributed environments. However, these capabilities also introduce significant security and privacy challenges that must be addressed at the system design level.

UAV systems are vulnerable to a range of cyber threats, including GPS spoofing, signal jamming, adversarial interference, and coordinated attacks targeting swarm communication protocols. Furthermore, drone operations over civilian and sensitive areas must comply with data protection and privacy regulations (e.g., GDPR, Ireland's Data Protection Acts, and China's PIPL), particularly when collecting telemetry, sensor readings, or image data. These constraints make traditional, centralized security solutions impractical or non-compliant.

Conventional Intrusion Detection Systems (IDS) typically rely on centralized machine learning models, which are unsuitable for UAV swarms due to intermittent connectivity, limited onboard computation, latency constraints, and the sensitivity of raw data. To address these limitations, Federated Learning (FL) enables collaborative model training across drones without requiring raw data to be shared. While promising, FL in UAV swarms introduces additional design challenges, including client dropout, heterogeneous hardware capabilities, malicious or unreliable participants, and the need for adaptive privacy guarantees.

In this project, students are expected to design and implement a federated IDS framework for UAV swarms that integrates Federated Learning with Reinforcement Learning (RL)-based trust modelling. The system should include an Adaptive Privacy Controller (APC) capable of dynamically adjusting privacy parameters (e.g., participation level, update frequency, or noise injection) based on real-time contextual factors such as threat level, mission criticality, trust scores, and resource availability.

Related work:

Wang, Xiaojie, Zhonghui Zhao, Ling Yi, Zhaolong Ning, Lei Guo, F. Richard Yu, and Song Guo. "A Survey on Security of UAV Swarm Networks: Attacks and Countermeasures." *ACM Computing Surveys* 57, no. 3 (2024): 1-37.

Husnoo, Muhammad Akbar, Adnan Anwar, Ripon K. Chakraborty, Robin Doss, and Mike J. Ryan. "Differential privacy for IoT-enabled critical infrastructure: A comprehensive survey." *IEEE access* 9 (2021): 153276-153304.

BDIC – BE IoT Project No. 6

Staff Member:	Dr. Diego Garaialde
Programme:	BE IoT
No. Students:	4
Project Title:	Design and Implementation of a Multi-Sensor IoT Platform for Gamified Physical Activity Monitoring

Project Description: Physical inactivity is a major global health risk, yet maintaining user motivation for regular exercise remains a significant challenge. While traditional fitness trackers record raw metrics (steps, heart rate), they often lack the psychological engagement required to sustain long-term behaviour change. "Exergames", described as video games that require physical exertion as a form of input, have emerged as a powerful solution to this problem. Notable examples, such as Pokémon GO, have demonstrated that location-based gameplay can significantly increase physical activity levels by integrating real-world movement with virtual rewards [2, 3].

This project aims to develop a modular IoT "Exergame" platform that bridges the gap between raw sensor data and user engagement. Students will design a system that ingests real-time data from various physical sensors (GPS, accelerometers, or wearable IoT nodes) and maps these inputs to a sophisticated gamification layer. The goal is to engineer a system where physical effort (such as walking speed, exercise repetitions, or distance travelled) is directly translated into game mechanics, validating the "proof of work" via sensor data analysis. This requires solving complex distributed system problems, such as synchronizing player state with a central server while minimizing latency and battery drain [4].

Suits students who like:

- Mobile Application Development (Android/iOS/Flutter)
- Distributed Systems & Network Architecture
- IoT Sensor Integration and Data Acquisition
- Game Design and Gamification Mechanics
- Signal Processing (filtering accelerometer/GPS data)
- UI/UX Design and Human-Computer Interaction

Project Overview: The primary objective of this project is to design and implement a scalable system that gamifies physical exercise. Students will work as a team to build a platform that collects data from heterogeneous sources, either from smartphone sensors to low-power IoT wearables, and processes this data to drive a real-time game environment. The project offers significant scope for technical exploration: from "pervasive sensing" challenges, such as optimizing client-side algorithms to filter noise and reduce battery drain before transmission [4], to system architecture, developing a robust backend that handles state synchronization and anti-cheating verification.

Beyond the technical implementation, the team is encouraged to explore different gamification strategies to maximize user engagement. The exact nature of the game, whether it is a territory capture game, a virtual pet simulator, or a competitive leaderboard system, is left to the students' creativity, provided it is

grounded in the principles of effective gamification [1]. The final deliverable should be a working prototype that demonstrates secure data transmission, accurate activity recognition, and a seamless user experience.

References:

[1] Hamari, J., Koivisto, J., & Sarsa, H. (2014, January). Does gamification work? A literature review of empirical studies on gamification. In 2014 47th Hawaii international conference on system sciences (pp. 3025-3034). IEEE.

[2] Lee, J. E., Zeng, N., Oh, Y., Lee, D., & Gao, Z. (2021). Effects of Pokémon GO on physical activity and psychological and social outcomes: a systematic review. *Journal of Clinical Medicine*, 10(9), 1860.

[3] Althoff, T., White, R. W., & Horvitz, E. (2016). Influence of Pokémon Go on physical activity: study and implications. *Journal of medical Internet research*, 18(12), e315.

[4] Shea, R., Fu, D., Sun, A., Cai, C., Ma, X., Fan, X., ... & Liu, J. (2017). Location-based augmented reality with pervasive smartphone sensors: Inside and beyond Pokemon Go!. *IEEE Access*, 5, 9619-9631.

BDIC – BE IoT Project No. 7

Staff Member:	Dr. Diego Garaialde
Programme:	BE IoT
No. Students:	4
Project Title:	IoT-Based Analysis of Circadian Rhythms and the Subjective Value of Mental Effort

Project Description: The willingness of the human brain to exert cognitive effort is not constant; it fluctuates significantly based on the body's internal clock (circadian rhythm) and sleep history. While the link between sleep loss and performance decline is well-documented, recent research highlights a subtler but critical impairment: a rise in the "Subjective Value of Mental Effort" (SVME). This means that fatigued individuals do not necessarily lose the ability to perform tasks, but rather lose the willingness to do so, viewing mental effort as significantly more "costly" [1, 2].

This project aims to build an IoT monitoring system to quantify this phenomenon. Students will develop a platform that combines passive sensing (wearable actigraphy and ambient light logging) to estimate circadian phase, with active testing (Ecological Momentary Assessment via a mobile app). The goal is to determine if disruptions in sleep or circadian alignment can predict a rise in "cognitive discounting"—the tendency to choose easier, less rewarding tasks over harder, more rewarding ones [3].

Suits students who like:

- Biomedical Signal Processing (Actigraphy, HRV analysis)
- Data Science & Time-Series Analysis
- Mobile Application Development (React Native/Flutter)
- Cognitive Science & Experimental Design
- Wearable Technology & API Integration (Fitbit/Smart Watch/Custom Hardware)

Project Overview: This project challenges students to create a "human-in-the-loop" IoT system that synchronizes physiological data with behavioral inputs. The team must design a data pipeline that continuously collects objective sensor data, such as wrist movement (accelerometry) and light exposure, to model the user's circadian rhythm and sleep efficiency [3]. Unlike standard sleep trackers, this system must process raw accelerometer data to extract "non-parametric circadian rhythms" (NPCRA), a method used in clinical research to quantify sleep fragmentation.

Simultaneously, the system must deploy "Cognitive Effort Discounting" tasks to the user's smartphone at algorithmically determined intervals. These tasks typically ask users to choose between a low-effort/low-reward option (e.g., "N-back level 1 for 1 point") and a high-effort/high-reward option ("N-back level 4 for 5 points"). The core scientific challenge lies in the data analysis: building a model that can identify the user's "peak" and "trough" hours and analyzing whether the cost of mental effort is significantly higher (i.e., users become more "risk-averse" regarding effort) during circadian dips.

References:

[1] Kool, W., & Botvinick, M. (2018). Mental labour. *Nature human behaviour*, 2(12), 899–908. <https://doi.org/10.1038/s41562-018-0401-9>

[2] Libedinsky, C., Massar, S. A., Ling, A., Chee, W., Huettel, S. A., & Chee, M. W. (2013). Sleep deprivation alters effort discounting but not delay discounting of monetary rewards. *Sleep*, 36(6), 899–904. <https://doi.org/10.5665/sleep.2720> [

3] Valdez P. (2019). Circadian Rhythms in Attention. *The Yale journal of biology and medicine*, 92(1), 81–92.

[4] Van Someren, E. J. W., Swaab, D. F., Colenda, C. C., Cohen, W., McCall, W. V., & Rosenquist, P. B. (1999). Bright Light Therapy: Improved Sensitivity to Its Effects on Rest-Activity Rhythms in Alzheimer Patients by Application of Nonparametric Methods. *Chronobiology International*, 16(4), 505–518. <https://doi.org/10.3109/07420529908998724>

BDIC – BE IoT Project No. 8

Staff Member:	Dr. Hadi Tabatabaee
Programme:	BE IoT
No. Students:	4
Project Title:	Green Autoscaling for LLM Inference in a VM-Based Mini-Cloud

Project Description: LLM inference is becoming a core building block in modern applications, but serving LLMs efficiently is challenging because demand is bursty and starting new workers can be slow (runtime initialisation and model loading). If the system reacts too late, users see long delays and timeouts; if it keeps too many workers ready “just in case,” it wastes compute, increasing energy use and carbon emissions. This project targets a publication-friendly question that goes beyond “predictive beats reactive”: how should an autoscaler decide when to pre-start workers when predictions are uncertain, and how can it minimise carbon while still meeting responsiveness targets?

You will build a desktop-based mini-cloud with several “nodes” represented as virtual machines (VMs). On top of this, you will implement a serverless-style inference service: requests arrive as events; the system can start/stop worker instances; and a controller decides how many workers should be active. LLM inference will use compact CPU-friendly models, so everything runs on a standard desktop. You will also build a workload generator that produces burst patterns including sudden spikes, periodic peaks, irregular bursts, and pattern shift (workload behaviour changes over time). The system will record request rate, queue length, response time, cold-start events, and resource usage to enable repeatable comparisons.

The core contribution is a confidence-aware, carbon-constrained autoscaling controller. You will implement two baselines: (1) a reactive controller (scale when load rises) and (2) a basic predictive controller (pre-start workers from a forecast). Your proposed controller will also forecast load, but crucially, it will estimate forecast confidence and use it to control how aggressively it pre-starts workers. When confidence is high, pre-starting avoids cold starts and protects tail latency; when confidence is low, the controller becomes conservative to avoid idle warm capacity and wasted carbon. Confidence will be implemented in an undergraduate-friendly way, such as recent forecast error (if the predictor has been wrong recently, trust it less) or agreement between two simple predictors (if they disagree, confidence is low). The “carbon-constrained” part means the controller is explicitly tuned to reduce carbon per successful request while keeping latency and timeout rates within agreed thresholds.

To study sustainability without specialised hardware, you will estimate energy from measured CPU time and memory footprint and combine it with time-varying grid carbon-intensity traces to compute energy-per-request and carbon-per-request. You will include a short sensitivity analysis (vary model parameters) to show the policy ranking remains stable. Evaluation will report p95/p99 latency, throughput, timeouts, cold-start counts, scaling stability, and a breakdown of overhead from cold starts versus idle capacity—alongside carbon/energy metrics.

Minimum expected outcome: VM mini-cloud; serverless-style LLM inference service with monitoring; bursty workload generator; reactive and predictive baselines; confidence-aware

carbon-constrained autoscaler; and an experimental study showing measurable carbon-efficiency gains without violating responsiveness targets. Extensions include carbon-intensity-aware thresholds, multi-tenant traffic classes, and comparisons of inference variants (e.g., smaller vs. larger or quantized models).

This is a build-and-evaluate group project that combines systems implementation with experimental evaluation. It is well-suited to a four-student team and can lead to a conference-style report, poster, or demo.

References

- [1] Zhu, S., Hurley, N., & Tabatabaee Malazi, H. “Does Energy-efficiency Mean Carbon-awareness? A Critical Evaluation of Sustainable Serverless Edge Scheduling.” UCC 2025.
- [2] Tari, M., et al. “Auto-scaling mechanisms in serverless computing: A comprehensive review.” Computer Science Review, 53 (2024): 100650.

BDIC – BE IoT Project No. 9

Staff Member:	Dr. Hadi Tabatabaee
Programme:	BE IoT
No. Students:	4
Project Title:	Prompt Engineering to Improve Intent-based Smart Home Reliability

Project Description: Smart environments should feel natural: a user says “make it warmer,” “turn off the lights,” or “reduce glare,” and the space responds safely and correctly. The SmartIntent project already provides an end-to-end intent-to-action pipeline for building automation, together with an execution/simulation harness. The gap is reliability. In practice, requests are incomplete (“make it cooler” — where, and by how much?), phrased in many different ways, and expressed across languages. LLM-generated plans can fail in predictable ways: invalid structured outputs, wrong device/room selection, unsupported actions, or “filling in” missing details by guessing. This project treats SmartIntent as a research platform and asks a publication-oriented question: which prompt-engineering strategies deliver notable reliability gains for intent-to-action automation, and why? The project is designed to be achievable within a semester by a four-student team, with clear milestones and reusable SmartIntent tooling.

You will not rebuild SmartIntent. Instead, you will improve its performance by designing prompts, prompting workflows, and evaluating them in controlled experiments using the SmartIntent codebase and execution harness (minor interface/integration adaptations are allowed; do not change the core intent-to-action logic). To make the system feel like a real user-facing IoT application, you will also connect a smartwatch as an input device. A lightweight watch app (or companion “quick action”) will send short intent messages to a gateway via HTTP or MQTT, optionally including simple context such as room selection or preset actions. This enables end-to-end demos and also introduces realistic, brief, “on-the-go” requests.

The project includes creating a small, high-quality multilingual benchmark dataset (e.g., English + Chinese) tailored to SmartIntent. To keep the workload manageable and consistent, build it around a fixed device catalogue and a small set of intent types (lighting/HVAC/blinds, etc.) with controlled context snapshots (rooms, available devices, optional sensor/time/occupancy fields). Each benchmark item includes the utterance, its context snapshot, a canonical reference label (intent type + required parameters, plus a reference action plan), and an ambiguity tag. An item is “ambiguous” when a required slot is missing (target room/device/value) or when constraints conflict; the correct behaviour is to ask for clarification rather than guess.

Starting from SmartIntent’s baseline prompting setup, implement a small set of reliability-focused prompt changes (e.g., higher schema/format compliance, stricter grounding to the provided catalogue/context, targeted clarification). Your evaluation must be controlled and explainable: compare baseline vs each change under identical conditions and include ablation-style comparisons. Validate the benchmark with schema checks, catalogue/rule checks (no unsupported devices/actions, parameter bounds), and a lightweight human audit of a subset.

Minimum expected outcome: smartwatch-to-SmartIntent input path; documented multilingual benchmark with scripts; baseline results; at least two prompt improvements with notable measurable gains; and a structured report in a conference-paper format (no submission required) explaining methods, ablations, and failure modes. Core requirements are well-defined; extension ideas (e.g., richer smartwatch UI, additional languages, more advanced prompting methods) are optional.

This is a build-and-evaluate group project that combines prompt engineering with rigorous benchmarking to improve the reliability of intent-based smart home automation. It is well-suited to a four-student team and can lead to a report, poster, or demo.

References

- [1] Shuyi Sun, et al. “SmartIntent: A Serverless LLM-Oriented Architecture for Intent-Driven Building Automation.” Proceedings of IEEE CloudCom, 2025.
- [2] Banghao Chen, et al. “Unleashing the Potential of Prompt Engineering in Large Language Models: A Comprehensive Review.” arXiv preprint arXiv:2310.14735, 2023.

BDIC – BE IoT Project No. 10

Staff Member:	Marco Alban-Paccha
Programme:	BE IoT
No. Students:	4
Project Title:	High-Throughput BLE 2M PHY Streaming of Multi-Channel Nerve Cuff Recordings

Project Description:

This project will design and evaluate a high-throughput Bluetooth Low Energy (BLE) 5 link, using the 2M PHY mode available on nRF52-class devices such as the Arduino Nano 33 BLE, for streaming multi-channel peripheral nerve cuff recordings. Students will be provided with nRF52-based hardware but will use only previously acquired wired datasets (supplied by the supervisor and/or from the literature) as the neural signal source; there will be no work with animals or real implants. The core question is whether a BLE 2M PHY link, configured with appropriate MTU, data length extension and connection interval, can support the sampling rates, channel counts and data quality required for state-of-the-art multichannel nerve cuff recordings that are typically processed with time-series and deep learning methods.

Students will implement firmware on the nRF52 hardware that replays the wired recordings from local storage as if they were being acquired in real time from an implanted nerve cuff front-end, packetising them for transmission over BLE 2M PHY. On the receiver side, a PC-based Python client or mobile app will act as BLE central, reconstructing the packet stream into continuous multichannel time series while logging throughput, packet loss, jitter and latency under a range of real wireless conditions (e.g. different distances, obstacles, interference). By comparing the reconstructed “wireless” data against the original wired recordings, the project will quantify how different BLE configurations and channel conditions affect signal-level metrics (SNR, waveform distortion) and downstream analysis such as compound action potential detection or CNN-based classification similar to recent multichannel cuff work. The final output will include detailed technical reports and configuration guidelines to support future development of fully implant-plus-node systems.

Suits students who like:

Wireless communications and IoT technologies

Embedded systems and protocol design

Signal processing and neural/bioelectric data

Python programming and applied data analysis

Experimental performance evaluation with real hardware

Project Overview:

The project is structured as a communications-and-signal-processing study, grounded in realistic nerve cuff data and exercised over real BLE hardware, but with no animal or invasive work. Students will first characterise the provided wired datasets (e.g. number of channels, sampling rate per channel, amplitude and noise statistics) and define a small set of representative recording “profiles” such as 4 channels at 10 kHz or 8 channels at 5 kHz, consistent with values reported for selective nerve cuff recordings. These profiles become the target operating points that the BLE link running on the nRF52 hardware must sustain with minimal packet loss and acceptable timing behaviour.

Next, students will develop and flash firmware to the provided nRF52 boards (e.g. Arduino Nano 33 BLE) to emulate a neural recording node. The device will read the stored multichannel time-series data and segment it into BLE packets according to configurable link parameters: PHY mode (2M), ATT MTU size, data length extension, connection interval, and number of notifications per interval, based on vendor guidance and community experience on achieving near-maximum throughput on Nordic SoCs. Using real hardware, students will then systematically test different wireless conditions by varying distance, orientation, obstacles and potential interference sources, recording how these factors impact achievable throughput and reliability compared with theoretical expectations and software-only simulations.

On the receiver side, a Python application (or alternative custom client) will connect to the nRF52 node as BLE central, receive notifications, reassemble samples into per-channel streams, and log detailed timing and error statistics. Using this toolchain, students will perform an experimental sweep over BLE parameters and physical conditions, measuring effective throughput (bytes/s), packet loss percentage, inter-packet interval distribution and end-to-end latency for each nerve-recording profile. The reconstructed signals will be compared to the original wired data using standard metrics such as SNR, root-mean-square error and timing error of detected events.

The outcome will be: (i) a set of recommended BLE 2M PHY configurations (MTU, data length, connection interval and buffering strategies) that can support specific multichannel nerve cuff recording profiles with near-wired quality on real hardware; and (ii) a detailed, well-structured technical report documenting methods, measurements, and design trade-offs to serve as a foundation for future development of fully integrated wireless neural interfaces using nRF52/54-class devices. This aligns with the programme’s focus on IoT and wireless communications while giving students hands-on experience with protocol-level

trade-offs, hardware constraints, and bioelectric signal requirements in a safe, dataset-driven environment.

Related Work:

Woodington BJ, Lei J, Carnicer-Lombarte A, Güemes-González A, Naegele TE, Hilton S, El-Hadwe S, Trivedi RA, Malliaras GG, Barone DG. Flexible circumferential bioelectronics to enable 360-degree recording and stimulation of the spinal cord. *Sci Adv.* 2024 May 10;10(19):ead11230. doi: 10.1126/sciadv.adl1230. Epub 2024 May 8. PMID: 38718109; PMCID: PMC11078185.

Carnicer-Lombarte A, Boys AJ, Güemes A, Gurke J, Velasco-Bosom S, Hilton S, Barone DG, Malliaras GG. Ultraconformable cuff implants for long-term bidirectional interfacing of peripheral nerves at sub-nerve resolutions. *Nat Commun.* 2024 Aug 30;15(1):7523. doi: 10.1038/s41467-024-51988-1. PMID: 39214981; PMCID: PMC11364531.

BDIC-BE IoT Project No. 11

Staff Member:	Xiulong Bao
Programme:	BE IoT
No. Students:	4
Project Title:	A Compact and Efficient Integrated Solar–RF Energy Harvesting System for IoT Applications

Project Description:

The increasing demand for sustainable and renewable energy sources has driven significant research into energy harvesting technologies. Harvesting energy from ambient sources, such as solar and RF, has become a promising solution to power small electronic devices, including sensors, IoT devices, and low-power electronics, without the need for batteries. While solar panels are widely used, their effectiveness is limited to daylight conditions. Although RF energy harvesting is less efficient than solar, it can be performed indoors and at night, thereby complementing solar energy harvesting. By combining solar power with RF energy harvesting, the system aims to provide a reliable and continuous power supply for IoT devices, which often operate in environments where energy sources can be intermittent. The proposed system will leverage advancements in both solar panel technology and RF energy conversion to ensure efficient energy capture and utilization. The ultimate goal is to enhance the sustainability and operational reliability of IoT devices by minimizing their reliance on conventional batteries and reducing maintenance needs.

Project Overview:

This project aims to develop a compact system that integrates solar and RF energy harvesting, providing a versatile and efficient solution for powering small electronic devices in various environments. The objectives include optimizing Energy Conversion Efficiency by improving the efficiency of both solar and RF energy harvesting to maximize power output. Additionally, the project will focus on developing a Smart Power Management Unit (PMU) that dynamically switches between solar and RF harvesting, optimizing energy collection based on environmental conditions. Finally, a prototype will be built and tested to validate the system's performance.

Expected Outcomes

- A prototype of a compact solar-RF harvesting system capable of efficient energy conversion.
- Demonstrated ability to provide a continuous power supply under varying environmental conditions.
- Publication of findings in relevant scientific conferences or journals.
- Potential for commercialization of the compact hybrid energy harvesting technology for various applications.

References

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- [2]. Buyun Yu, Zhehan Wang, Lu Ju, Chao Zhang, Zhenguo Liu, Li Tao, and Weibing Lu, Flexible and Wearable Hybrid RF and Solar Energy Harvesting System, IEEE Transactions on Antennas and Propagation, Vol. 70, No.30, 2022, pp.2223-2233.

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